

Disassemble bonded composite structure using shock waves generated by a flashlamp-pumped pulsed laser

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1. INTRODUCTION

This paper describes a technique of using pulse laser to disassemble a portion of bonded composite structure. Using this method, patches on a composite structure can be cleanly and simply removed instead of using mechanical methods such as milling or grinding. Detail of the experiment will be discussed to illustrate the potential as well as the challenges of this technique.

2. EXPERIMENT

The experiment was performed with the purpose of making debonding in bonded plate. In the following, each component of the experiment is described in more detail.

A. Laser

The laser used in this test is a flash-pumped pulsed laser with the wavelength of 1064 nm. The laser has pulse width from 200 μ s to 10 ms and power from 2.4 J to 55 J as shown in Fig. 1.

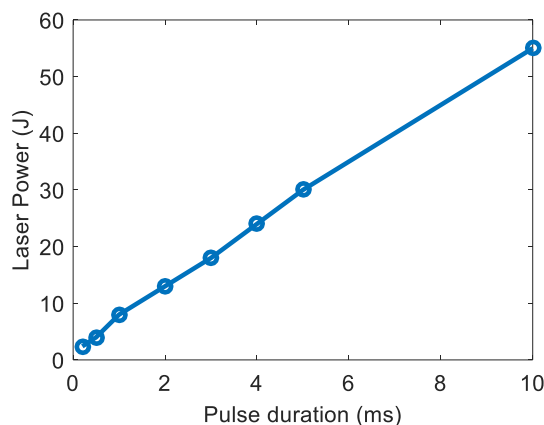


Fig.1 Laser power vs. pulse width

B. Composite plate with bonded patch

A portion of a 1-mm thickness composite plate (Toray T300/#2500 carbon/epoxy prepreg [0/90]_{2s}) was cut and bonded to the same plate using epoxy glue (see Fig. 2). The bond was compressed and let dried in room temperature. Also showed in Fig. 2 is a thin copper tape as the sacrificed layer for the laser shock experiment. The sketch of the cross-section of the composite plate with bonded patch and applying the pulse laser is showed in Fig. 3.

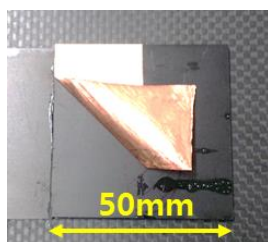


Fig.2 A 2-mm thickness bonded specimen

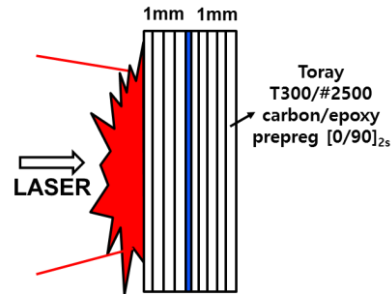


Fig. 3. Cross-section of the bonded composite plate with patch

C. Experiment setup

The experiment set up is shown in Fig. 4. The laser was at the distance of 100 mm from the target. We can see again in this figure the thin copper tape as the sacrificed layer on the specimen. The laser beam coming out of the laser aperture has the diameter of 6 mm and focus to the specimen using a focus lens with focal length of 50 mm. The location of the lens was adjusted so that the laser beam diameter on the specimen was approximately 4 mm. This diameter was measured using the “burn paper” technique.

The laser was set at maximum laser power, which is 55 J corresponding to the pulse width of 10 ms. One single pulse was applied to the specimen. We observed that when the diameter was about 4 mm, only the scarified layer was damaged while the specimen was still intact.

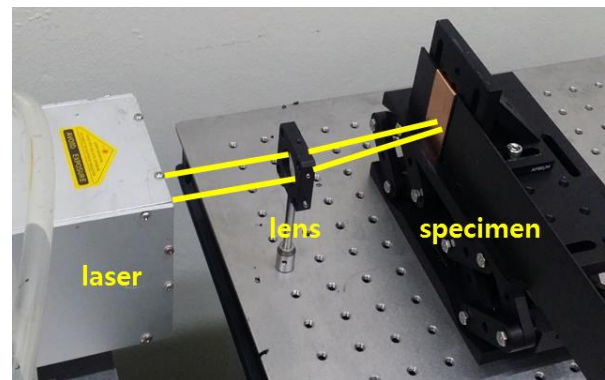


Fig. 4 Experiment setup: the pulse laser is focused on to the composite plate with bonded patch

Figure 5 shows the principle of the laser shock technique described in this paper [1]. A shock wave is generated by the pulse laser. When the initial shock wave reaches to the back of the bonded specimen, a reflected wave is generated. This reflection wave, when crossing with the initial wave generates a strong tensile force and makes the debonding between bonded layers. It is possible that the bonding between laminates in the composite plate or in the patch also

debonded, however since usually the bonding between layers is much weaker than the laminate bonding, the debond damage is more likely to occur at the bonding layer.

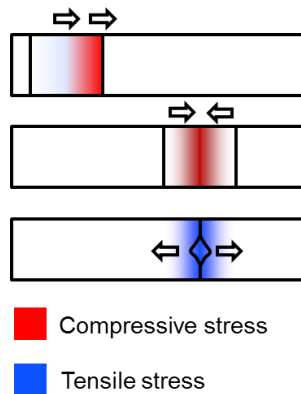


Fig. 5 Laser shock technique principle

D. Results

Figure 6 shows the C-scan of the specimen before and after the laser shock experiment. We can see that at location of the laser impact, a debonding damage was made. The depth of the debonding damage is 1.16 mm under the top surface, which evidently showed that the debonding is between two bonded layers.

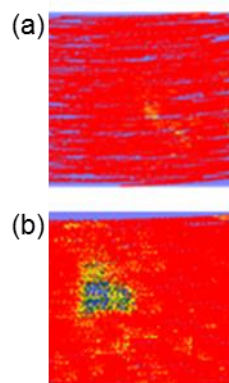


Fig. 6 C-scan comparison (a) before and (b) after the laser shock experiment

E. Experiment with a 4-mm thickness bonded composite specimen

In the following, we increase the thickness of the bonded specimen to see if the laser shock experiment described above able to successfully make the debonding. Fig. 7 shows a 2-mm thickness CFRP wing-box specimen with two patches (both are 1-mm thickness Toray T300/#2500 carbon/epoxy prepreg $[0/90]_{2s}$), which composed a total 4-mm thickness bonded specimen. The C-scan test was performed before and after the laser shock experiment to check if the debonding was made. However, for this case, we could not find any clear evidence of the debonding damage.

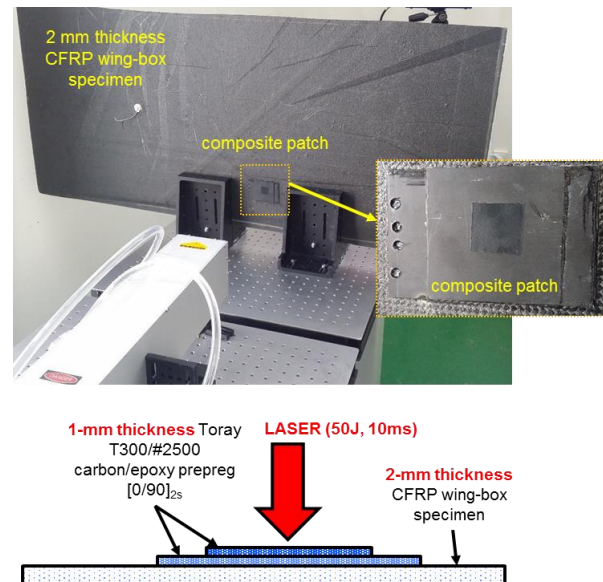


Fig. 7 A 4-mm thickness bonded specimen

3. DISCUSSIONS

This study reports the early effort of using pulse laser to disassemble a portion of bonded composite structure. We did success to make damage of a thin specimen with the thickness of 2-mm. However, when the specimen is thicker (4-mm), we failed to achieve the debonding.

The solution can be found by studying several approaches. One obvious solution is to increase the power and reduce the pulse width of the laser. Secondly, we may use two identical lasers applied to the specimen at two opposite sides with a small time gap between two pulses. This approach is attractive in the sense that we can well control the location of the crossing waves as well as increase the amplitudes of two crossing waves. Other minor solution may involve using confined materials such as water or glass to increase the power of initial shock wave into the specimen [2].

One application of this study is for removing the unwanted parts (such as patches) on the aircraft structures. This technique promises a quicker and simpler method than the existing techniques currently in use.

REFERENCES

- 1) Romain E., Michel B., Fabienne T., Frederic P., Laurent B., Laurence C.A., Bastien E., Clemens B., A study of composite material damage induced by laser shock waves. Composites : Part A, 53, 54-64, 2013
- 2) Fabbro R. , Fournier J., Ballard P., Devaux D., and Virmont J., Physical study of laser-produced plasma in confined geography, J of Appl Phys, 68, 775, 1990